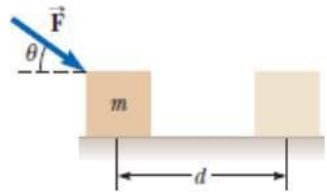
Sheet (3) Work and Energy

1) Suppose you push on a block sliding on a table. Your push has a magnitude of 50 N and makes a downward angle of 60° with the direction of motion. What is the work you do on the block while the block moves 1.6 m?

2) A block of mass m = 2.50 kg is pushed a distance $d = 2.2\ m$ along a frictionless, horizontal table by a constant applied force of magnitude F = 16.0 N directed at an angle $\theta = 25.08^{o}$ below the horizontal as shown in the Figure. **Determine** the work done on the block by each of the following force



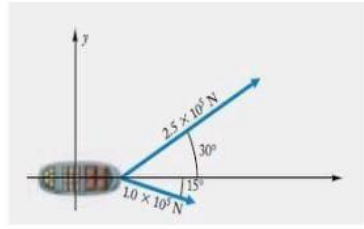
(a) The applied force

(b) The normal force exerted by the table

(c) The gravitational force

(d) The total work on the block.

3) Consider the barge being pulled by two tugboats, as described in Figure 3.2. The pull of the first tugboat is $2.5 * 10^5\ N$ at 30° to the left, and the pull of the second tugboat is $1.0 * 10^5\ N$ at 15° to the right (see Figure). What is the work done by each tugboat on the barge while the barge moves 100 m forward (in the direction of the x-axis in the Figure? What is the total work done by both tugboats on the barge?

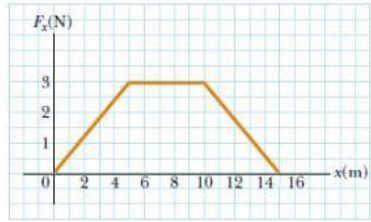


4) A 7.80-g bullet moving at 575 m/s strikes the hand of a superhero, causing the hand to move 5.5 cm in the direction of the bullet's velocity before stopping. Find

(a) **The force** and **the work done by that force** to stop the bullet

(b) **The time** between the moment the bullet strikes the hand and the moment it stops

5) A particle is subjected to a force F_x that varies with position as shown in the figure. Find the work done on the force when it moves :



(a) From $x = 0$ to $x = 5m$

(b) From $x = 5$ to $10\ m$

(c) From $x = 10$ to $15\ m$

(d) The total work done by the force from $x = 0$ to $x = 15m$.

6) When a 4.00-kg object is hung vertically on a certain light spring that obeys Hooke's law, the spring stretches 2.50 cm. If the 4 kg object is removed.

(a) How far will the spring stretch if a 1.50-kg block is hung on it, and

(b) How much work must an external agent do to stretch the same spring 4.00 cm from its unstretched position?

7) A 0.600-kg particle has a speed of 2.00 m/s at point A and kinetic energy of 7.50 J at point B. **Determine:**

(a) its kinetic energy at A

(b) its speed at B,

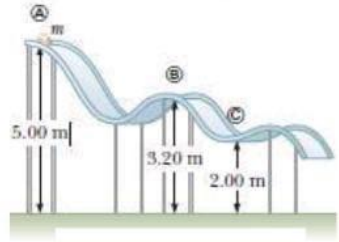
(c) The net work done on the particle by external forces as it moves from A to B?

8) Surplus energy from an electric power plant can be temporarily stored as gravitational energy by using this surplus energy to pump water from a river into a reservoir at some altitude above the level of the river. If the reservoir is 250 m above the level of the river, how much water (in cubic meters) must we pump to store $2.0 * 10^{13}\ J$? (Giving that the water density is 1000Kg/m3)

9) Because of brake failure, a bicycle with its rider careens down a steep hill 45 m high. If the bicycle starts from rest and there is no friction, what is the final speed attained at the bottom of the hill?

10) A roller coaster begins at rest from a first peak, descends a vertical distance of 45 m, and then ascends a second peak, cresting the peak with a speed of 15 m/s. How high is the second peak? Neglect the friction.

A block of mass m = 5.0 kg is released from point A and slides on the frictionless track shown in the Figure. Determine



(a) The block's speed at points B and C

(b) The net work done by the gravitational force on the block as it moves from point A to point C.